

# AYMANE AIT DADS

AI Research Intern | LLM Fine-Tuning · Computer Vision · Ablation Studies

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## PROFESSIONAL SUMMARY

Achieved top 10% public leaderboard (0.80/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Kaggle Competition by fine-tuning a 30B Mamba-Transformer MoE model with LoRA, conducting 20+ single-variable ablation studies to isolate optimal training configurations. Ranked 1st in EURECOM class on an AI-generated image detection benchmark (0.791 AUC) using CLIP ViT-L/14 fine-tuning, with results submitted to the NTIRE 2026 @ CVPR CodaBench international benchmark. Core competencies span LLM fine-tuning (LoRA, PEFT, SFTTrainer), computer vision (ViT, CLIP, CNN), reproducible research methodology, and cross-disciplinary applied AI experimentation. Available for a 6-month internship from 2026, based in France (EU academic visa), with full authorization to work across Europe.

## CORE COMPETENCIES

LLM Fine-Tuning | Reproducible Research | Ablation Studies | Computer Vision | Applied AI Research | Model Architecture | Academic Benchmarks | Cross-disciplinary Collaboration

## WORK EXPERIENCE

### Orange Maroc

Jul 2024 - Aug 2024

**Data Science Intern** - Casablanca, Morocco

- Designed and evaluated a clustering pipeline combining Random Forest and K-Means on **100K+ fiber-optic network records**, mapping customer profiles to 5 commercial service tiers for network optimization.
- Reduced manual analysis time by **~60%** through automated feature engineering and model evaluation workflows applied to upload speed, latency, and jitter signals.
- Delivered stakeholder presentations translating **applied AI model outputs** into actionable network maintenance recommendations consumed directly by engineering teams.
- Built end-to-end data science pipeline in Python (Pandas, Scikit-learn) and visualized results in **Power BI**, enabling reproducible reporting across thousands of network nodes.

## PROJECTS

### NVIDIA Nemotron Reasoning Challenge

Kaggle Competition · In Progress (June 2026) · Top 10% Public Leaderboard

- Achieved **top 10% public leaderboard score (0.80/1.0)** in a \$106K+ prize-pool competition by fine-tuning Nemotron-3-Nano-30B, a hybrid Mamba-Transformer MoE (30B total / 3.5B active parameters, 52 layers), using a LoRA adapter (r=32, ~888M trainable parameters) trained on 7,828 chain-of-thought reasoning examples across 6 logic puzzle types.
- Resolved **5 Blackwell GPU (sm\_120) incompatibilities** in strict sequential order - including monkey-patched allocator warmup, stubbed mamba3/cutlass imports, and a pure PyTorch RMSNorm fallback - and conducted 20+ single-variable ablation studies confirming that completion-only loss and all-linear LoRA targeting were each independently critical to achieving peak score.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton

### AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · 1st in Class

- Ranked **1st on EURECOM private Kaggle class leaderboard (0.791 AUC)** by fine-tuning CLIP ViT-L/14 (428M parameters, pretrained on 400M image-text pairs) with a custom MLP head and dual learning-rate strategy across 250K training images from 25+ generative model types.
- Submitted independently to the **NTIRE 2026 @ CVPR CodaBench** international benchmark; confirmed via ablation that 1-epoch training (0.793 AUC) outperforms 4-epoch (0.767 AUC), demonstrating superior out-of-distribution generalization through 10-view TTA ensemble.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · Kaggle GPU

## EDUCATION

### Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Image Security, Statistics, Cloud Computing

### CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

**SKILLS**

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**LLM Fine-Tuning & NLP:** LoRA / QLoRA | SFTTrainer | PEFT | Hugging Face Transformers | Mamba-Transformer MoE | vLLM | Chain-of-Thought

**Computer Vision & Deep Learning:** PyTorch | CLIP / ViT | CNN / DenseNet | OpenCLIP | TensorFlow | Scikit-learn | FP16 Mixed Precision

**Research Engineering & MLOps:** Ablation Studies | CUDA Debugging | Triton | Git / Linux | Python | Pandas / NumPy | Power BI